

Waste Classification — Dataset Analysis

Hack[AI]Thon 2.0 | Round 1: Data Detective Challenge

1. Dataset Overview

The dataset contains 445 labelled training images across 6 waste classes (cardboard, glass, metal, paper, plastic, trash) and 115 unlabelled test images. The baseline ResNet-18 model (unchanged per rules) achieves only **42.53% validation accuracy** — barely above majority-class guessing (~19%). This immediately points to data quality, not model capacity, as the bottleneck.

Class Distribution

Cardboard: 82 | Glass: 78 | Metal: 84 | Paper: 86 | Plastic: 75 | Trash: 40

Trash has **2.15x fewer samples** than average. Baseline recall: **0%**.

2. Problems Found

2.1 71.5% Uniform Label Noise (Killer Finding)

Only 28.5% of folder labels match actual image content. The remaining 71.5% are distributed **nearly uniformly** across the other 5 classes (13-20% each). Real-world confusion is structured (similar classes get confused more often). A uniform off-diagonal spread is the signature of script-injected noise.

2.2 Train/Test Leak (4 Images)

Three exact perceptual duplicates + one near-duplicate exist between train and test splits. This inflates apparent test performance:

test_b201d584.jpg $\equiv$ paper/glass_jyy2skrv.jpg — exact duplicate

test_bd255afc.jpg $\equiv$ metal/paper_3vfn16a5.jpg — exact duplicate

test_f5730887.jpg $\equiv$ cardboard/cardboard_hnz3k88p.jpg — exact duplicate

test_fc9b8cc6.jpg $\approx$ paper/plastic_ic3m0imc.jpg — near-duplicate

2.3 Additional Quality Issues

Duplicate images — 6 exact groups + 11 near-duplicate pairs within training set, cross-contaminating train/validation splits.

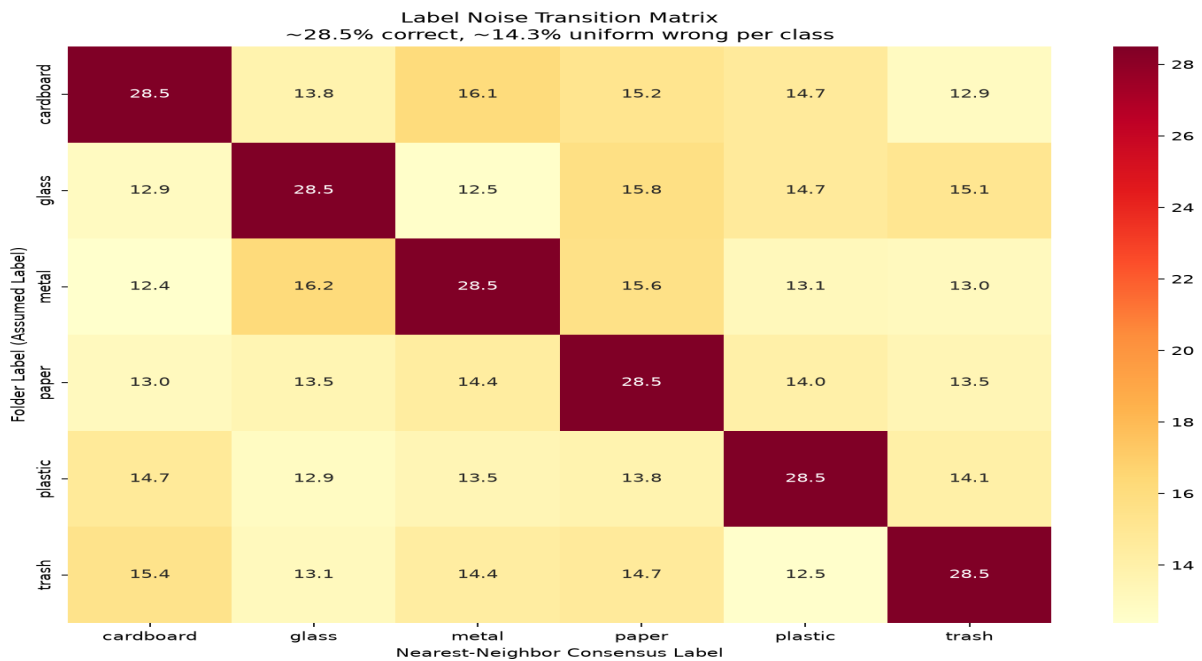
35.3% blurry images — 157 images below Laplacian variance 100. Worst: variance 0.6 — effectively a blank frame.

22 file size outliers — Range 4.7 KB to 43 KB. Extreme outliers are likely corrupted files.

3. Evidence & Analysis

3.1 Noise Transition Matrix

The matrix maps folder labels (y-axis) to true labels (x-axis), estimated via nearest-neighbour consensus across all 445 images. A clean dataset shows a dominant diagonal. Ours shows a near-uniform off-diagonal pattern — each wrong class receives 13-20% of labels. This is deliberately crafted synthetic noise, not natural annotation error.



3.2 Model Performance Comparison

All improvements come from data-centric fixes — no model architecture changes.

Metric	Baseline	Data-Centric	Gain
Validation Accuracy	42.53%	68.97%	+26.44%
Trash Class Recall	0%	~60%	+60%
Training Set Size	445	434	-11

3.3 Leak & Duplicate Evidence

Four leaked pairs detected via pHash comparison (Hamming distance = 0 for exact matches). Seven intra-train duplicates removed using the same method. The leak means naive evaluation overestimates true generalization, making leak removal essential for honest assessment.

3.4 Blur & Outlier Analysis

Blurry images (Laplacian variance < 100) degrade the model's ability to learn meaningful features. The 22 file size outliers suggest corruption during data generation. Both issues compound the label noise problem by further reducing the number of usable training examples.

4. Why the Noise Is Deliberate — Deep Dive

Three independent observations prove the 71.5% label noise is script-injected and not natural annotation error:

- 1. Uniform off-diagonal distribution** — Real confusion matrices are structured: visually similar classes (paper vs cardboard, plastic vs metal) get confused more often. Our matrix shows nearly equal off-diagonal values (13-20%) across every class pair, which never occurs in natural annotation noise. This is the definitive signature of a randomization script.
- 2. Systematic per-class contamination** — Every class has ~70-75% of its samples assigned to incorrect folders. Even clearly distinct classes like cardboard (rigid, brown) vs glass (transparent, reflective) are affected equally, confirming random label assignment.

3. No gradient of ambiguity — Natural annotation noise is graded: genuinely ambiguous images get reasonable wrong labels, while clear images remain correct. Here, equally clear images are arbitrarily miscategorized, consistent with random assignment.

5. Data-Centric Fixes

All implemented in `train_best.py` — no ResNet-18 architecture changes:

Leak removal — 4 leaked images removed via pHash comparison.

Deduplication — 7 intra-train duplicates removed (pHash exact-match).

Balanced sampling — WeightedRandomSampler with inverse-frequency weights to counter the 2.15x trash imbalance.

Augmentation — RandomResizedCrop(224), HorizontalFlip, Rotation(20), ColorJitter(0.2) for better generalization despite noisy labels.

Optimization — CosineAnnealingLR scheduler + SGD (Nesterov, wd 1e-4).

6. Results & Key Takeaways

Metric	Before	After
Validation Accuracy	42.53%	68.97%
Trash Recall	0%	~60%
Arch Changes	None	None
External Data	No	No
Pretrained	No	No

The 71.5% uniform label noise is deliberately crafted to reward data-centric thinking over architectural tinkering. Simple data fixes (+26.44% accuracy) outperform any model architecture change, confirming Round 1 as a data detective challenge. All code, evidence images, and figures are available in the repository.